

Data Engineering

Technical Challenge

Medallion Architecture · Databricks · LLM Integration

Deadline	5 business days from receipt
Platform	Databricks Free Edition
Dataset	electronics_dataset.xlsx (provided)
Delivery	Live demo on Databricks Free Edition + verbal walkthrough

Objective

Build a Medallion Architecture pipeline in Databricks that ingests the provided electronics dataset, applies data quality and normalization steps, uses an LLM to parse product description into structured format, validates the results with an LLM-as-Judge pattern, and surfaces a clean analytical gold table.

Pipeline Overview

#	Stage	Description
1	Ingest (Bronze)	Load the provided Excel file and write two raw tables.
2	Standardize (Silver)	Normalize both tables: (weight → kg, price → USD float), resolve duplicate vendor names.
3	LLM Extraction	Create a tasks that uses an LLM to extract Name, Sub-Category, and Brand from the Description column. Pass the allowed category list as context. Allowed Sub-Category values (pass in the prompt): Televisions, Computers, Accessories, Phones, Smartwatches
4	LLM Judge	Create a task with an LLM Judge that compares previous task output against the approved taxonomy. Records that fail are flagged and is stored for manual review. Approved Taxonomy: Televisions, Computers, Accessories, Phones, Smartwatches, Printers, Cameras, Consoles, Hardware
5	Taxonomy (Silver)	Store LLM successful outputs enriched with Vendor info.
6	Aggregation (Gold)	Produce Aggregated Table with: Category, Sub-Category, # Products, AVG Price, Min Price, Max Price.

Technical Rules

- Use Databricks Free Edition: <https://www.databricks.com/learn/free-edition>
 - All tables must follow the Medallion Architecture.
 - Pipeline should handle Schema evolution.
 - LLM choice is open: open-source models or an external API (OpenAI, Anthropic Claude, Google Gemini).
 - LLM extraction accuracy will not be evaluated — only the logic and implementation quality.
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Evaluation Criteria

The candidate will present their solution live, including:

- A working end-to-end pipeline running on Databricks Free Edition.
- Explanation of design decisions and architecture choices.
- Discussion of challenges encountered and how they were resolved.

Evaluation will focus on: code quality, pipeline logic, correct use of the Medallion Architecture, and the LLM integration approach — not on the accuracy of LLM outputs.

Bonus Points

The following items are optional but will be considered in the overall evaluation:

★ Bonus Criteria

Version Control & CI/CD - Demonstrative

Publish your code to a repository (GitHub, Bitbucket, or Azure DevOps) and include a basic demonstrative CI/CD pipeline (e.g., lint, unit tests, or notebook validation on push).

Databricks Asset Bundles (DABs) - Demonstrative

Package and deploy the pipeline using Databricks Asset Bundles for a reproducible, environment-agnostic deployment.

Data Quality Expectations

Apply QA techniques such as Great Expectations or Databricks built-in expectations at key pipeline stages.

MLflow Traceability

Instrument all LLM calls with MLflow for traceability — log prompts, responses, latency, and flag decisions.

Idempotency

Ensure the pipeline can be triggered multiple times without duplicating data.

Prompt Versioning

Version-control the prompts used in both the LLM extraction and LLM Judge steps, storing them in MLflow or a config file so any change is traceable and reproducible.